Software Requirements Specification (SRS)

Automatic Attendance System Using Face Recognition

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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to define the functional and non-functional requirements of the **Automatic Attendance System Using Face Recognition**. This document provides a complete description of the system, its features, constraints, interfaces, and expected behavior.

The system aims to automate attendance marking using facial recognition technology, reducing manual effort, preventing proxy attendance, and improving the accuracy and efficiency of attendance management.

This document serves as a foundation for:

- System design and development
- Validation and testing
- Communication among stakeholders
- Future maintenance and upgrades

1.2 Document Conventions

The following conventions are used in this document:

- **System** refers to the Automatic Attendance System.
- **Admin** refers to authorized users such as teachers or administrators.
- **User** refers to students or employees whose attendance is recorded.
- Functional requirements are labeled as **FR**.
- Non-functional requirements are labeled as **NFR**.
- Use cases are identified as **UC**.

Requirement Priority Levels:

- High – Essential requirement
- Medium – Important but not critical
- Low – Optional enhancement

1.3 Intended Audience

This document is intended for:

- Project Supervisors
- Developers
- Testers
- System Analysts
- Educational Institutions or Organizations using the system
- Future Maintenance Teams

1.4 Product Scope

The scope of this project includes:

- Registration of individuals with facial data.
- Real-time face detection and recognition.
- Automatic attendance marking.
- Attendance record storage and retrieval.
- Administrative management of users.
- Attendance report generation.
- Secure and accurate attendance tracking.

The system is targeted for:

- Universities and colleges
- Schools
- Offices
- Training institutes

1.5 References

References used for system understanding and development:

- IEEE SRS Documentation Standards
- OpenCV Documentation
- Python Documentation
- Face Recognition Library Documentation
- SQLite Documentation

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2. Overall Description

2.1 Product Perspective

The Automatic Attendance System is a standalone application that replaces traditional attendance methods.

Traditional System Problems:

- Time consuming
- Human error prone
- Proxy attendance possible
- Difficult record maintenance

Proposed Solution:

- Contactless attendance marking
- Automated facial identification
- Database-driven attendance records
- Report generation and analytics

The system consists of:

1. Face Recognition Module
2. Attendance Management Module
3. Admin Control Module
4. Database Management Module

2.2 Product Features

Major features include:

Feature 1: User Registration

- Register new students/employees.
- Capture facial dataset.
- Store identity information.

Feature 2: Face Recognition Attendance

- Real-time face detection.
- Match captured face with trained dataset.
- Mark attendance automatically.

Feature 3: Admin Management

- Add user
- Update user
- Delete user
- View user records

Feature 4: Attendance Reports

- Daily attendance reports
- Monthly reports
- Export records
- Search attendance logs

Feature 5: Security and Authentication

- Admin login authentication
- Secure database access
- Restricted administrative operations

2.3 User Classes and Characteristics

Admin

Characteristics:

- Basic computer knowledge
- Authorized access
- Manages attendance data

Responsibilities:

- Register users

- Train facial data
- Monitor attendance
- Generate reports

Student/Employee

Characteristics:

- Registered in system
- Facial data enrolled
- Uses system for attendance

Responsibilities:

- Present face for recognition

2.4 Operating Environment

Hardware Requirements:

- Computer/Laptop
- Webcam or IP Camera
- Minimum 4GB RAM
- Processor Core i3 or above

Software Requirements:

- Windows/Linux
- Python 3.x
- OpenCV
- Face Recognition Library
- SQLite
- Tkinter or Browser (HTML/CSS interface)
- VS Code

2.5 Design and Implementation Constraints

Constraints include:

- System accuracy depends on camera quality.
- Poor lighting may affect recognition.
- Limited performance on low-end hardware.
- SQLite may limit scalability for large institutions.
- Internet dependency may apply if cloud deployment is used.

2.6 Assumptions and Dependencies

Assumptions:

- Users are properly registered.
- Camera functions correctly.
- Adequate lighting available.
- Face recognition model is trained.

Dependencies:

- OpenCV libraries
- Face recognition algorithms
- SQLite database
- Python runtime environment

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3. System Features and Requirements

3.1 Functional Requirements

FR-1 User Registration

Description:
System shall allow admin to register new users.

Inputs:

- Name
- ID
- Department/Class
- Facial images

Process:

- Capture face images
- Generate facial dataset
- Store data in database

Output:

- Successful registration confirmation

Priority:
High

FR-2 Face Detection

Description:
System shall detect faces through webcam in real time.

Input:
Live video stream

Output:
Detected facial region

Priority:
High

FR-3 Face Recognition

Description:
System shall compare detected faces with trained data.

Process:

- Extract facial features
- Match against stored templates
- Identify person

Output:
Recognized identity

Priority:
High

FR-4 Automatic Attendance Marking

Description:
System shall mark attendance automatically once recognition is successful.

Attendance record shall include:

- User ID
- Name
- Date
- Time
- Status (Present)

Priority:
High

FR-5 Duplicate Prevention

Description:
System shall prevent multiple attendance entries for same user on same day.

Priority:
High

FR-6 Attendance Report Generation

Description:
System shall generate:

- Daily reports
- Weekly reports
- Monthly reports

Priority:
Medium

FR-7 Search Attendance Records

Description:
Admin can search attendance records using:

- Name
- ID
- Date

Priority:
Medium

FR-8 User Management

System shall allow:

- Add user
- Edit user
- Delete user

Priority:
High

FR-9 Admin Login

Description:
System shall authenticate admin before allowing access.

Inputs:

- Username
- Password

Priority:
High

FR-10 Data Export

Description:
System shall export attendance reports to:

- CSV
- Excel
- PDF (optional)

Priority:
Medium

3.2 Non-Functional Requirements

NFR-1 Performance

- Face recognition response under 3 seconds.
- System should support multiple users efficiently.

NFR-2 Accuracy

- Recognition accuracy should exceed 90%.
- Minimize false positives and false negatives.

NFR-3 Security

- Password-protected admin panel.
- Encrypted database access where applicable.
- Restricted access for unauthorized users.

NFR-4 Reliability

- System should maintain attendance records without corruption.
- Recovery should be possible after failures.

NFR-5 Usability

- User-friendly interface.
- Easy navigation for admin.
- Simple attendance operations.

NFR-6 Scalability

- System should support increasing number of users.
- Future migration to MySQL possible.

NFR-7 Maintainability

- Modular code structure.
- Easy debugging and updates.

3.3 External Interface Requirements

User Interface Requirements

Interface includes:

Admin Dashboard:

- Login Screen
- Registration Form
- Attendance Dashboard
- Report Panel

User Interface:

- Camera Capture Window
- Recognition Display
- Attendance Confirmation

Hardware Interfaces

- Webcam for image capture
- Computer system

Software Interfaces

- Python Backend
- OpenCV Library
- SQLite Database
- Tkinter GUI or HTML/CSS frontend

Communication Interfaces

- Internal communication between:
    - Recognition module
    - Database module
    - Attendance module

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4. Data Requirements

4.1 Data Entities

Main entities:

User Entity

Attributes:

- User_ID
- Name
- Department
- Facial Data

Attendance Entity

Attributes:

- Attendance_ID
- User_ID
- Date
- Time
- Status

Admin Entity

Attributes:

- Admin_ID
- Username
- Password

Relationship Overview

- One user can have many attendance records.
- Admin manages many users.

Sample Database Tables

Students Table

Field	Type
Student_ID	Integer
Name	Text
Department	Text
Face_Data	BLOB

Attendance Table

Field	Type

Field	Type
Attendance_ID	Integer
Student_ID	Integer
Date	Date
Time	Time
Status	Text

4.2 Data Integrity and Security

Integrity Requirements:

- No duplicate attendance entries.
- Referential integrity between user and attendance tables.
- Input validation for all forms.

Security Requirements:

- Password authentication.
- Secure storage of credentials.
- Controlled admin access.
- Backup mechanism for records.

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5. Use Case Model

5.1 Use Case Diagram

Main Actors:

1. Admin
2. Student/Employee

Use Cases:

Admin:

- Login
- Register User
- Train Dataset
- Manage Records
- View Attendance

- Generate Reports

Student:

- Present Face
- Mark Attendance

Use Case Diagram Representation:

```
        +--------------------+
        |      Admin         |
        +--------------------+
            |   |   |   |
            |   |   |   |
  --------------------------------------
  | Login                              |
  | Register User                      |
  | Manage Records                     |
  | View Reports                       |
  --------------------------------------

        +--------------------+
        | Student/Employee   |
        +--------------------+
                |
        -----------------
        | Mark Attendance|
        -----------------
```

5.2 Use Case Descriptions

UC-1 Admin Login

Actor:
Admin

Precondition:
System running

Flow:

1. Admin enters credentials.
2. System validates login.
3. Dashboard opens.

Postcondition:
Admin authenticated.

UC-2 Register User

Actor:
Admin

Flow:

1. Enter user information.
2. Capture facial images.
3. Store dataset.

Postcondition:
User registered.

UC-3 Mark Attendance

Actor:
Student

Flow:

1. Face appears before camera.
2. System detects face.
3. Face recognized.
4. Attendance marked.

Postcondition:
Attendance stored.

UC-4 Generate Reports

Actor:
Admin

Flow:

1. Select report type.
2. System retrieves records.
3. Report generated.

Postcondition:
Attendance report available.

UC-5 Manage Records

Actor:
Admin

Flow:

1. Search user.
2. Edit/Delete data.
3. Save changes.

Postcondition:
Records updated.

Appendices

Appendix A. Glossary

Term	Meaning
Face Recognition	Identifying a person through facial features
OpenCV	Computer vision library
Dataset	Collection of facial training images
SQLite	Lightweight relational database
Attendance Record	Stored presence information

Appendix B. Analysis Models

Context Diagram (Textual)

Inputs:

- Facial Images
- User Information

- Admin Commands

System Processes:

- Face Recognition
- Attendance Processing
- Report Generation

Outputs:

- Attendance Records
- Reports
- Notifications

Data Flow Model (Level 0)

```
User Face Input
     |
Face Detection
     |
Face Recognition
     |
Attendance Processing
     |
Database Storage
     |
Reports Output
```

Future Enhancements (Optional)

- Cloud database integration
- Mobile app support
- Mask-aware recognition
- SMS/Email attendance alerts
- Multi-camera support

Conclusion

This Software Requirements Specification defines the requirements for developing the Automatic Attendance System Using Face Recognition. The proposed system provides a secure, automated, and efficient attendance solution while minimizing manual effort, improving accuracy, and preventing proxy attendance.